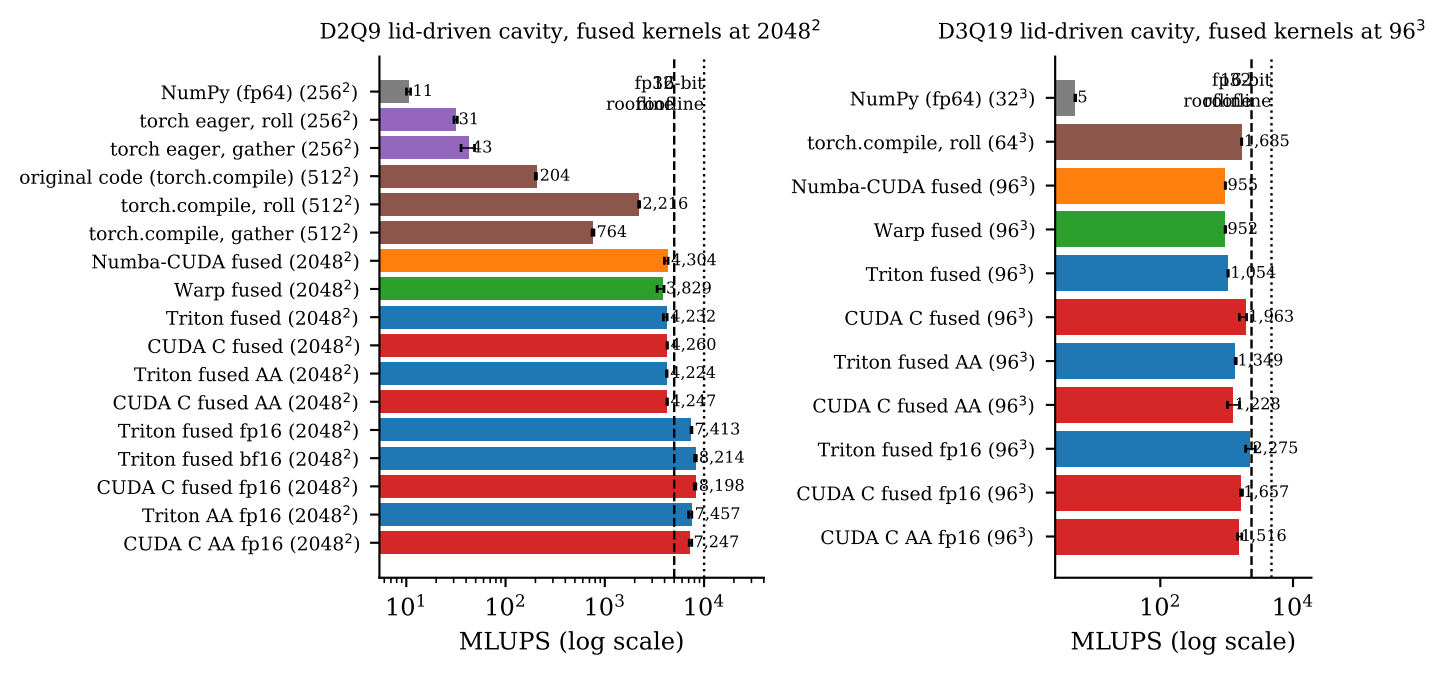
D2Q9 lid-driven cavity, fused kernels at 2048²
NumPy (fp64) (256²)
torch eager, roll (256²)
torch eager, gather (256²)
original code (torch.compile) (512²)
torch.compile, roll (512²)
torch.compile, gather (512²)
Numba-CUDA fused (2048²)
Warp fused (2048²)
Triton fused (2048²)
CUDA C fused (2048²)
Triton fused AA (2048²)
CUDA C fused AA (2048²)
Triton fused fp16 (2048²)
Triton fused bf16 (2048²)
CUDA C fused fp16 (2048²)
Triton AA fp16 (2048²)
CUDA C AA fp16 (2048²)
11
31
43
204
2,216
764
4,304
3,829
4,232
4,260
4,224
4,247
7,413
8,214
8,198
7,457
7,247
fp32 roofline
fp16 roofline
10^1
10^2
10^3
10^4
MLUPS (log scale)
D3Q19 lid-driven cavity, fused kernels at 96³
NumPy (fp64) (32³)
torch.compile, roll (64³)
Numba-CUDA fused (96³)
Warp fused (96³)
Triton fused (96³)
CUDA C fused (96³)
Triton fused AA (96³)
CUDA C fused AA (96³)
Triton fused fp16 (96³)
CUDA C fused fp16 (96³)
CUDA C AA fp16 (96³)
5
1,685
955
952
1,054
1,963
1,349
1,228
2,275
1,657
1,516
fp32 roofline
fp16 roofline
10^2
10^4
MLUPS (log scale)